

How Ethos and Pathos Appeals Resonate in Reader Interpretations of Social Media Messages

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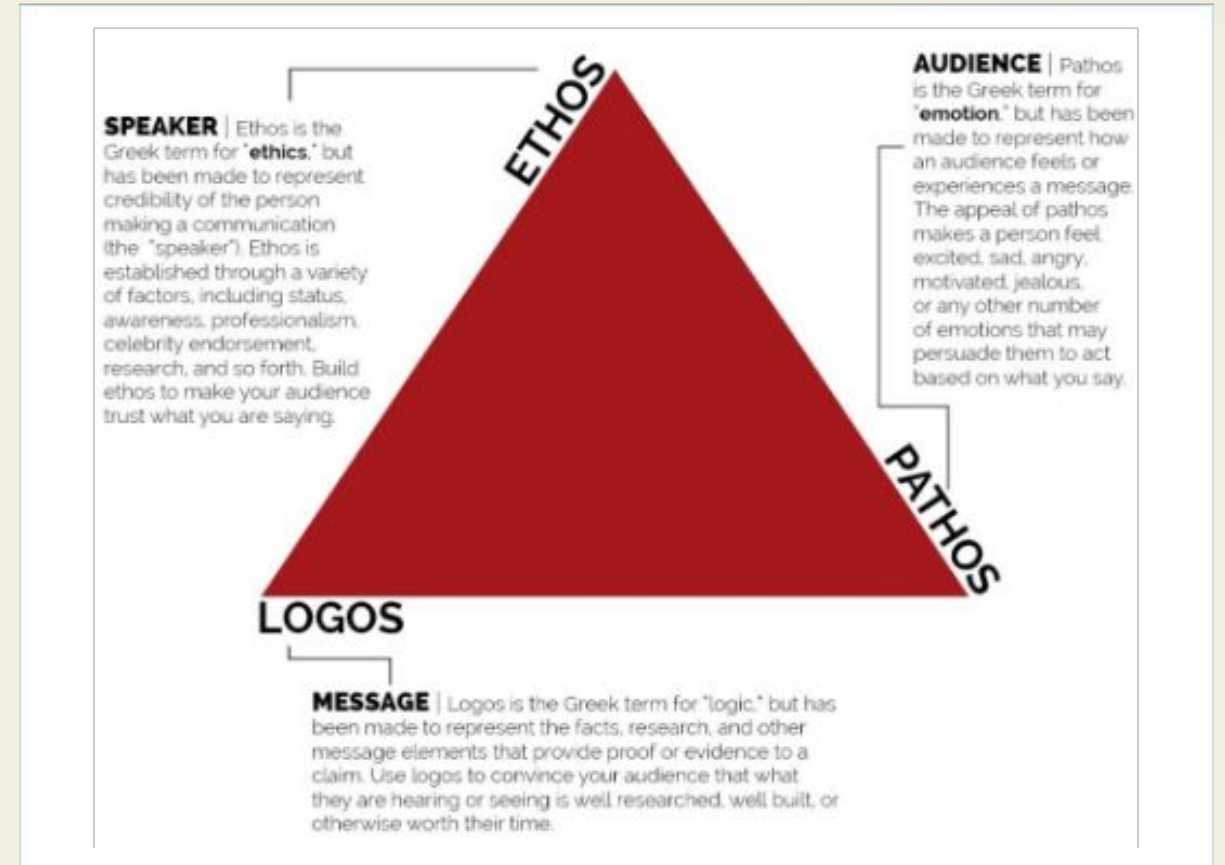
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Motivation: Aristotle's Means of Persuasion

Modes of Rhetoric

- ✓ **Ethos:** Credibility and moral character of the speaker.
- ✓ **Pathos:** Emotional appeals targeting the audience's affective state.
- ✓ Both directly target interpersonal trust and orientation rather than static truths.



Motivation: Reader Interpretation

The Universal Audience

90% of **social media** users consume content without commenting.

However, they remain **rhetorically present**: they observe, internalize, and shape discourse internally.

We study the *rhetorical situation of interpretation* rather than visible participation in social media.



The Research Gap



Explicit Bias

The focus on *visible* strategies in posts so far ignores the silent receiver's **internal representation**.



Transformation

We lack empirical evidence of how appeals are **retained or omitted** as they move from sender to reader.



Interpretation

NLP needs to pivot from modeling what text says to modeling the distribution of what text **evokes**.

Investigating Rhetorical Resonance

- ✓ **RQ1: Resonance Alignment:** Are ethos and pathos in the original sentence reflected consistently in audience interpretations?
 - ✓ **RQ2: Trigger Variability:** Which specific rhetorical labels trigger the greatest variability in audience understanding?
 - ✓ **RQ3: Attitude Prediction:** Can these rhetorical appeals predict a reader's attitude toward the author?
 - ✓ **RQ4: Latent Factors:** Which socio-cognitive factors beyond rhetoric drive interpretive divergence?
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Methodology: Dataset and Approach



OrigamIM Dataset

Corpus: 2,018 sentences from Reddit r/ChangeMyView.
9,851 human-written interpretations + Attitude scores.

Silver-Standard Labels:

Finetuned RoBERTa-large (Ethos) & RoBERTa-base (Pathos).
Gemini 2.5 with Zero-shot, One-shot and CoT prompting.
Final labels via majority voting.

Validation Approach:

Data analysis with regression models.

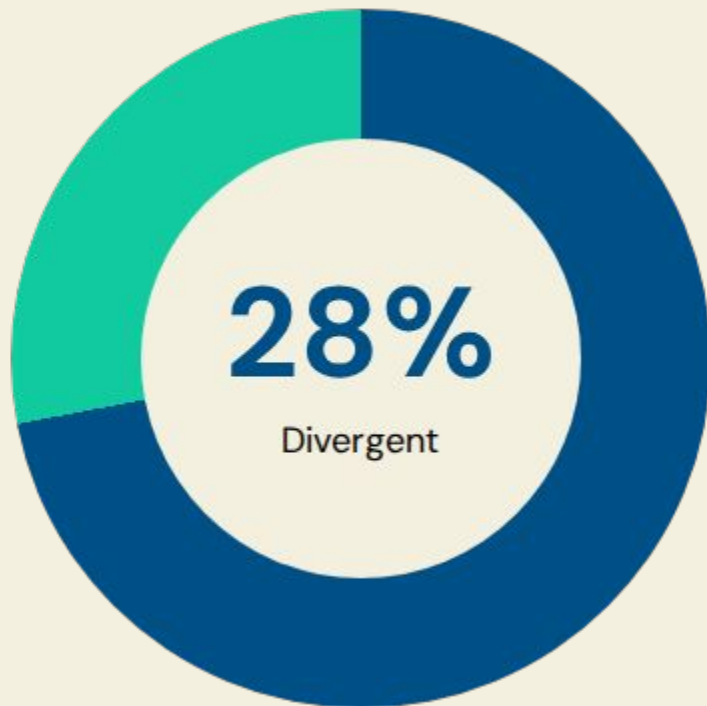
Results: Classification Performance

Recognition of Ethos and Pathos in Social Media Messages and Interpretations



Gemini 2.5 shows higher recall for ethos cues, while both models achieve comparable performance for emotional appeals.

Validation: RQ1. The Resonance Gap



Interpretive Variability

A significant chasm exists between textual content and audience reception.

- 72% Overall Label Alignment
- 28% Divergent Interpretations

Full Alignment across all interpretations of the original comment occurs in only **16%** of comments.

| Validation: RQ2. The Charge of Ambiguity

Rhetorically charged content amplifies disagreement

1.1–1.5x

Variability Multiplier

- ✓ **Neutrality Fosters Consensus:** Neutral sentences show significantly higher alignment ($p < 0.001$).
- ✓ **Positive Pathos ($\beta = 0.30$):** Significantly associated with increased interpretive variability.
- ✓ **Positive Ethos ($\beta = 0.46$):** Credibility reinforcement triggers the highest divergence in internal representation.

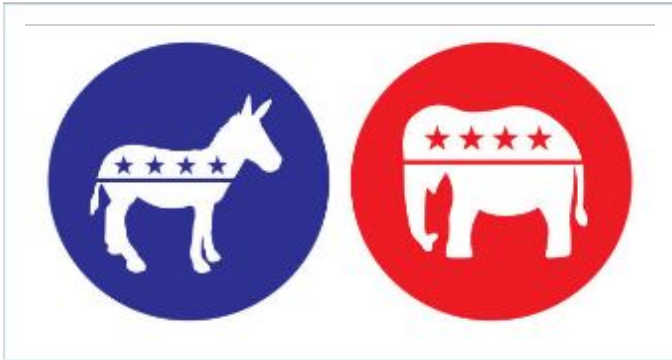
Validation: RQ3. Predicting Audience Attitudes

Rhetorical appeals are significant predictors of reader perception of the author

Predictor (Source Appeal)	Coefficient (B)	Std. Error	t-value	Significance
Positive Pathos [+]	0.37	0.04	8.89	*** (p < .001)
Negative Pathos [-]	-0.12	0.02	-5.78	*** (p < .001)
Positive Ethos [+]	0.18	0.04	4.62	*** (p < .001)
Negative Ethos [-]	-0.10	0.02	-4.46	*** (p < .001)

Note: Positive appeals elicit more positive attitudes; negative appeals trigger more negative perception.
Pathos has a higher effect size.

Validation: RQ4. Drivers of Divergence



Interpreter-related factors

Arise from the background, beliefs, and perspectives of the interpreter rather than from the sentence itself.



Form-based factors

Driven by linguistic or rhetorical properties of text that invite multiple interpretations (sarcasm, 1st vs 3rd person framings).



Contextual factors

Stem from missing or underspecified contextual information (e.g. very short sentences, ambiguous referents).

Conclusions and Future Work

Readers **do not interpret** the same message in the same way.

Rhetorical strategies **increase interpretive variability**, particularly in polarized contexts.

Meaning is **co-constructed** by the audience.

NLP should move **beyond text-centric** representations towards receiver-centric models.



Questions?

Thank you for your attention.

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